
C2G-BENCH: A Cyber-Physical Evaluation Benchmark for Hierarchical Reinforcement Learning in Grid-Interactive Hyperscale Data Centers

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Abstract

1 Hyperscale data centres now draw 250 MW or more. At this scale, they can pro-
2 vide grid frequency regulation as ancillary services. Testing whether RL agents
3 can safely manage both sub-second hardware control and 15-minute market bid-
4 ding requires a benchmark that captures the full cyber-physical loop. No such
5 benchmark exists.

6 We introduce C2G-BENCH, a benchmark for evaluating hierarchical RL agents
7 in grid-interactive data centres under FERC Order 755 regulation. C2G-BENCH
8 provides two Gymnasium environments at coupled timescales: C2GFastEnv
9 (5-second ticks, 4-dimensional action over DVFS, CDU pump, HVAC, and a
10 150 MWh BESS) and C2GMacroEnv (15-minute market commitments wrapping
11 180 sub-steps). Seven physics engines simulate a 250 MW two-zone facility using
12 analytical solutions, not numerical ODE solvers. Real Alibaba workload traces
13 and NOAA weather data drive the simulation. Six ISO market configurations
14 span three continents. Four stress scenarios target specific failure modes: GenAI
15 demand spikes, near-limit ambient temperature, and compound battery-cooling
16 faults.

17 C2G-BENCH evaluates agents on five metrics: tracking RMSE, thermal violation
18 rate, throughput ratio, BESS degradation, and survival rate. Reporting all five
19 is necessary because reducing them to a single score hides important trade-offs
20 between grid service quality, safety, and compute output.

21 Baseline results show that SAC reaches 476 kW tracking RMSE on the default
22 scenario, but PPO fails with 0% survival under compound stress. Rule-based
23 agents survive all scenarios but track poorly (14–18k kW RMSE). These results
24 define open evaluation problems for hierarchical RL research. All results are
25 simulation-only; hardware-in-the-loop validation is future work.

26 1 Introduction

27 Generative AI has increased the power demand of hyperscale data centres to 250 MW or more [?].
28 This level of consumption places significant stress on regional transmission grids, especially when
29 combined with volatile renewable generation. Grid operators need these large loads to transition
30 from passive consumers to active assets that provide ancillary services such as frequency regula-

tion. This transition is important because the continued scaling of AI infrastructure depends on the capacity and stability of the energy system.

Previous work has shown that data centers can provide frequency regulation through computational flexibility. Wang et al. [2019] demonstrated that DVFS on CPU dummy loads can track a regulation signal, but their method wastes power and does not model thermal constraints or modern GPU workloads. ? used chilled water thermal storage to participate in energy and regulation markets, but relied on classical MPC that struggles with the non-stationary bursts of GenAI serving. ? reviewed AI-driven Virtual Power Plants and identified the autonomous coordination of heterogeneous assets as an open challenge.

The core difficulty is the interaction between two timescales. Grid regulation signals arrive every 2 to 4 seconds, while market commitments span 15 minutes and AI training jobs run for hours. Existing simulators focus on either macro-level energy costs or low-level distribution physics, but they do not capture the coupling between sub-second hardware control and market-level bidding under realistic safety constraints.

Why this matters for evaluation. Without a shared benchmark, claims about HRL performance in this domain cannot be reproduced or compared. A policy that achieves low tracking error may do so by exhausting the battery, and this trade-off is invisible unless the evaluation captures multiple metrics simultaneously. C2G-BENCH fills this gap by providing a complete evaluation substrate: physics simulators, market models, stress scenarios, a non-reducible metric suite, and reference baselines – all under a single reproducible framework.

We present C2G-BENCH, a cyber-physical evaluation benchmark with the following contributions:

- **Benchmark environment.** Two Gymnasium environments (C2GFastEnv at 5 s, C2GMacroEnv at 15 min) co-simulating a 250 MW data centre with six ISO market configurations spanning the US, Europe, and Oceania. Observations are 17-dimensional and the seven-component reward reflects measurable grid-service and hardware-safety objectives.
- **High-fidelity simulators.** Seven physics engines (workload, thermal, electrical, BESS, macro-grid, renewable, weather) using exact-exponential analytical solutions, driven by real Alibaba traces (v2023–v2026) and NOAA weather records.
- **Safety shield.** A Simplex-architecture safety filter that enforces five hard constraints (thermal, BESS SoC, frequency, voltage) in $\mathcal{O}(1)$ time, applicable to any agent at evaluation time.
- **Evaluation suite.** Four rule-based and learned baselines (Random, Rule-Based MPC, PPO, SAC) evaluated on a five-metric non-reducible suite across four stress scenarios. SAC achieves a tracking RMSE of 476 kW on the default scenario, while PPO and all baselines fail on the Thermal Squeeze and Battery Drain scenarios, establishing concrete targets for future HRL research.

2 Related Work

Data centre energy management. ? showed that model-predictive control reduces cooling energy in Google data centres by 30%, but did not model grid regulation or BESS. Wang et al. [2019] used DVFS on dummy loads to track a frequency signal; this wastes power and does not model thermal constraints or GenAI workloads. ? added chilled water storage for regulation market participation, but used classical MPC that fails under the bursty, heavy-tailed load patterns of modern GenAI serving. ? reviewed AI-driven VPPs and identified autonomous coordination of heterogeneous assets as an open problem.

Grid-interactive buildings. Virtual Power Plants aggregate distributed flexibility resources, but existing VPP simulators do not model GPU thermal dynamics or BESS degradation at the fidelity needed for sub-5-minute regulation tracking.

RL benchmarks. OpenAI Gym [?] and Gymnasium [?] established the standard interface we adopt. Unlike robotics or game benchmarks, C2G-BENCH evaluates agents on safety-critical infrastructure where thermal violations have real consequences and the reward function maps directly to measurable economic and grid outcomes.

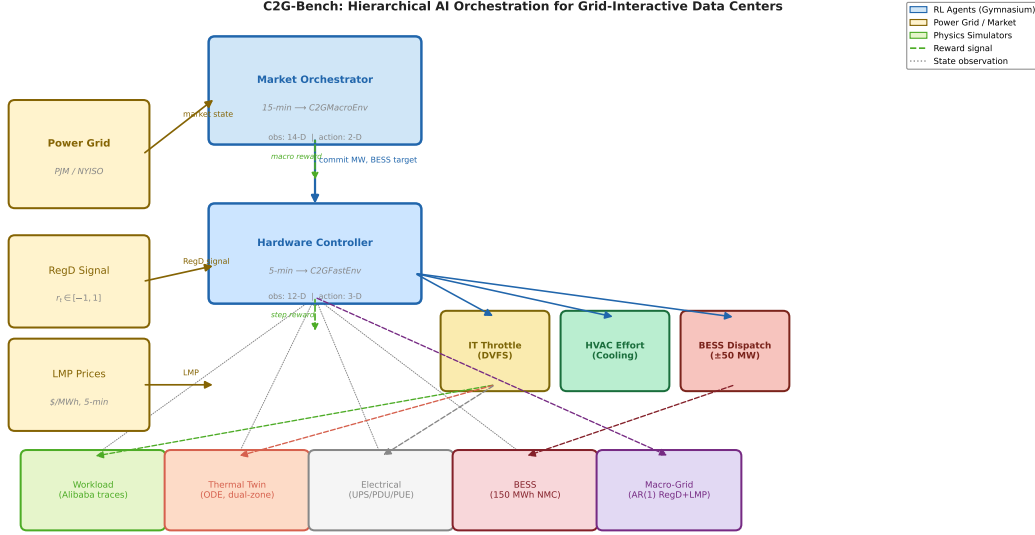


Figure 1: C2G-BENCH hierarchical architecture. C2GMacroEnv sets 15-minute market commitments; C2GFastEnv executes 5-second hardware control across four physical levers and seven physics simulators.

80 **What is missing.** None of the above provides a complete evaluation substrate that combines:
81 (a) sub-5-second hardware control with 15-minute market bidding, (b) multiple ISO market con-
82 figurations for cross-market generalisation studies, (c) a non-reducible metric suite that exposes
83 hidden trade-offs, and (d) stress scenarios that systematically expose failure modes. C2G-BENCH
84 addresses all four gaps.

85 3 The C2G-BENCH Environment

86 3.1 Physical Simulators

87 C2G-BENCH integrates seven independent, unit-tested physics engines running at a canonical
88 timestep of $\Delta t = 5$ s. Each engine exposes a stateless `step()` method; the Gym environments
89 compose them at every tick. All temporal dynamics use exact-exponential analytical solutions; no
90 numerical ODE solvers are used.

91 **Workload Orchestrator.** Models a 250 MW two-zone facility: 2,000 liquid-cooled GPU racks
92 (Zone A) and 2,500 air-cooled CPU racks (Zone B). Rack power follows a superlinear utilisation
93 model:

$$P_{\text{rack}}(u) = P_{\text{idle}} + (P_{\text{max}} - P_{\text{idle}}) u^{\alpha}, \quad (1)$$

94 with $\alpha_A = 1.4$ (GPU) and $\alpha_B = 1.2$ (CPU). The three traces map to racks as follows: batch v2023
95 drives 1,200 Zone A racks (P_{flex} , throttleable via DVFS); GenAI v2026 drives 800 Zone A racks
96 ($P_{\text{base,A}}$, rigid); DLRM v2025 drives 2,500 Zone B racks ($P_{\text{base,B}}$, rigid). 74% of total IT power is
97 rigid; the agent controls only the remaining 26% through batch throttling. Unserved batch work
98 defers into a FIFO queue tracked via Little’s Law.

99 **Thermal Twin.** Models Zone A and Zone B as independent lumped-capacitance thermal masses
100 with exact-exponential integration:

$$T_z(t+\Delta t) = T_{\text{eq}} + (T_z(t) - T_{\text{eq}}) e^{-K_{\text{tot}} \Delta t / C_z}, \quad (2)$$

101 where T_{eq} is the equilibrium temperature given current heat input, cooling, and ambient coupling.

102 Zone A uses liquid cooling with a CDU pump speed action $\omega_{\text{pump}} \in [0.15, 1]$ that controls effective
103 conductance. At minimum pump speed, the thermal time constant extends from ≈ 12.7 min to

104 ≈ 85 min, allowing the water loop to act as a short-horizon thermal battery. Zone B uses air cooling
 105 with HVAC fan effort $u_{\text{hvac}} \in [0, 1]$. Both zones share a safety limit of $T_{\text{safe}} = 35^\circ\text{C}$ (episode
 106 termination) and warning threshold $T_{\text{warn}} = 33^\circ\text{C}$ (reward penalty).

107 **Electrical and Loss Model.** Maps IT + cooling power to facility draw through three closed-form
 108 loss stages: UPS (double-conversion, load-dependent efficiency), PDU (fixed 1.5–2.0%), and main
 109 transformer (iron + copper losses at 300 MVA). Dynamic PUE is computed every tick as $\text{PUE}(t) =$
 110 $P_{\text{facility}}/P_{\text{IT}}$.

111 **BESS Simulator.** A 150 MWh / 50 MW NMC Li-ion battery with round-trip efficiency that de-
 112 grades with C-rate and SOC:

$$\eta(C, \text{SOC}) = \max(0.70, \eta_{\text{peak}}^2 - k_C C^2 - k_{\text{SOC}}(\text{SOC} - 0.5)^2), \quad (3)$$

113 with $\eta_{\text{peak}} = 0.97$, $k_C = 0.008$, $k_{\text{SOC}} = 0.010$. SOC is bounded to $[0.10, 0.95]$ with smooth power
 114 derating near limits. Capacity fade accumulates linearly with throughput.

115 **Macro-Grid Signal Generator.** Produces two exogenous signals per tick, parameterised by a mar-
 116 ket preset (Section 3.3). The regulation signal is an AR(1) process: $r_t = \rho r_{t-1} + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$,
 117 with market-specific ρ and σ . A drift correction enforces energy neutrality over the settlement win-
 118 drow. The LMP uses a load-duration proxy calibrated per market.

119 **Renewable Generator.** Models 100 MW wind (IEC power curve) and 75 MW solar PV (PVUSA
 120 model) with degradation, driven by the weather engine.

121 **Weather Engine.** Loads real NOAA ISD data or generates calibrated synthetic temperatures using
 122 a two-harmonic model with per-market parameters (Table 2). Ambient temperature drives cooling
 123 COP and envelope heat gain.

124 3.2 Data

125 C2G-BENCH is driven by open-access datasets preprocessed into aligned 5-minute-resolution
 126 CSVs:

- 127 • **Alibaba Batch v2023** (`batch_v2023.csv`). 33 days of GPU scheduling data [?]. We extract
 128 `gpu_milli_request` and normalise by the 99th percentile (12,620 milli-GPU) to get per-tick
 129 utilisation u_{batch} . 78% of ticks have zero arrivals, making this trace highly bursty.
- 130 • **Alibaba DLRM v2025** (`dlrm_v2025.csv`). 30 days of DLRM inference serving [?]. We extract
 131 `active_gpu_count` and normalise by 227 to get u_{dlrm} . This trace is continuous and predictable,
 132 with a clear diurnal pattern.
- 133 • **Alibaba GenAI v2026** (`genai_v2026.csv`). 288 ticks (one day) of LLM inference traffic, tiled
 134 cyclically. Ticks with `avg_gpu_duty_cycle` above the 75th percentile (12.19%) are flagged as
 135 spikes (1_{spike}). About 25% of ticks are in spike mode.
- 136 • **Energy load data.** Real zonal load from six ISOs (NYISO, PJM, CAISO, ERCOT, ENTSO-E,
 137 AEMO), converted to LMP through a load-duration proxy calibrated per market.
- 138 • **NOAA weather.** Hourly dry-bulb temperature from co-located weather stations [?]. For markets
 139 without local data, a calibrated two-harmonic synthetic model reproduces the correct seasonal and
 140 diurnal envelope.

141 All datasets are bundled under their original open-access licences. Preprocessing scripts are in
 142 `preprocessing/`.

143 3.3 Supported Grid Markets

144 C2G-BENCH supports six ISO market configurations spanning three continents (Table 1). Each
 145 preset encodes a distinct regulation product, AR(1) signal statistics, LMP calibration, and settlement
 146 window. The physical simulators, observation spaces, and reward functions are identical across
 147 markets, enabling direct cross-market policy transfer studies.

C2G-Bench Simulator Suite — Fidelity Validation

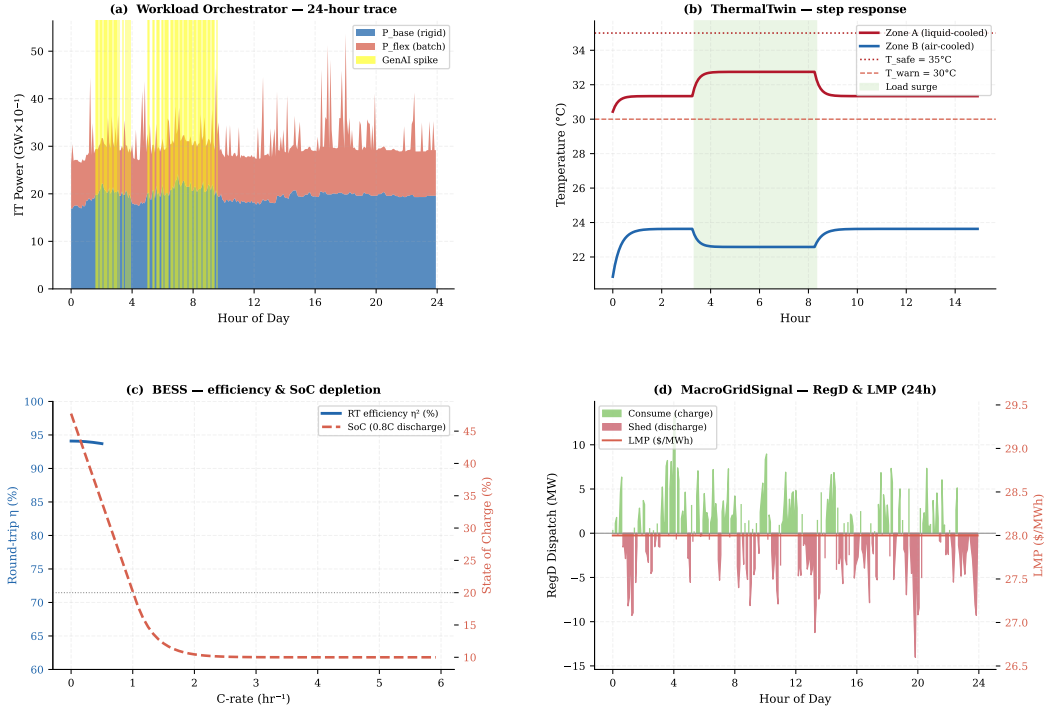


Figure 2: Simulator fidelity. (a) 24-hour IT power trace showing rigid P_{base} and throttleable P_{flex} with GenAI spikes. (b) Zone A and B step-response under a load surge. (c) BESS efficiency vs C-rate and SOC depletion at 0.8C. (d) RegD dispatch with NYISO LMP overlay.

Table 1: Supported grid market presets. ρ and σ are AR(1) parameters at the 5-second timestep. LMP base and slope in USD/MWh.

ID	ISO	Hub	Reg. product	ρ	σ	Base	Slope	Archetype
nyiso_nyc	NYISO	New York City	AGC Secondary	0.9963	0.022	28	18	Baseline (real data)
pjm_dom	PJM	Ashburn VA	RegD fast AGC	0.9963	0.022	35	10	Largest market
caiso_pgae	CAISO	Bay Area	REGU / REGD	0.9960	0.025	48	28	Duck curve
ercot_north	ERCOT	Dallas-Fort Worth	ECRS	0.9950	0.030	28	55	Isolated grid
entso_de	ENTSO-E	Frankfurt	FCR-N	0.9980	0.015	60	8	Deep EU market
aemo_nsw	AEMO	Western Sydney	Regulation FCAS	0.9945	0.032	55	45	High renewables

3.4 Gym Environments

C2G-BENCH provides two Gymnasium environments [?] at complementary timescales, connected in a hierarchical control stack (Figure 1). Observations are normalised to approximately $[-1, 1]$ or $[0, 1]$; actions are clipped to their declared boxes before passing to simulators.

Hierarchical Architecture

The two environments form a two-level hierarchy in which C2GMacroEnv acts as a *grid manager* and C2GFastEnv acts as a *hardware controller*:

1. C2GMacroEnv observes 15-minute aggregated statistics, executes a market bid, and sets the regulation commitment c_{MW}^* for the inner environment.
2. C2GFastEnv receives c_{MW}^* and operates the four physical levers (DVFS, CDU pump, HVAC, BESS) at 5-second resolution to track the committed signal.

Table 2: Weather calibration per market. SH = southern hemisphere (phase negated). Src: R = real CSV, S = calibrated synthetic.

Market	Station	$\bar{T}$	A_{ann}	A_{day}	Range	Src
nyiso_nyc	Central Park	13.0	12.5	5.0	$[-20, 42]$	R
pjm_dom	Reagan Natl	15.0	13.0	7.0	$[-15, 42]$	S
caiso_pgae	SJC Mineta	14.5	6.5	8.0	$[-5, 42]$	S
ercot_north	DFW Intl	19.5	15.0	9.0	$[-15, 46]$	S
entso_de	Frankfurt FRA	11.0	10.5	7.0	$[-20, 40]$	S
aemo_nsw	Bankstown SH	18.5	9.0	8.0	$[-5, 46]$	S+SH

Coupling is implemented via a property setter: C2GMacroEnv writes the cleared commitment to C2GFastEnv.committed_mw before unrolling 180 sub-steps. For research into hierarchical decompositions, C2GMacroEnv exposes an inner_action_fn callback: if provided, the function is called at every sub-step as $a_{\text{low}} = f(\text{obs}_{\text{low}}, a_{\text{macro}})$, enabling a trained low-level policy to replace the safe defaults. Without the callback, the hardware levers are held at conservative fixed values ($\theta=1$, $\omega_{\text{pump}}=0.7$, $u_{\text{hvac}}=0.7$, $d_{\text{bess}}=0$) so that C2GMacroEnv is usable as a stand-alone market-bidding environment.

C2GFastEnv – Low-Level (5-second). Observation space (17-D Box): normalised Zone A and B temperatures (T_A/T_{safe} , T_B/T_{safe}), BESS state-of-charge, base and flex IT load fractions, facility power, signed regulation signal $r \in [-1, 1]$, normalised LMP, grid load, GenAI spike flag $\mathbf{1}_{\text{spike}}$, previous throttle and pump actions, PUE, ambient temperature, frequency deviation, PCC voltage, and batch queue backlog (capped at $2 \times P_{\text{flex,max}}$).

Action space (4-D Box):

1. DVFS throttle on batch workloads $\theta \in [0, 1]$ (0 = full shed, 1 = full throughput). Controls the 26% of facility power that is schedulable.
2. CDU pump speed $\omega_{\text{pump}} \in [0, 1]$ (Zone A liquid loop). At minimum speed the thermal time constant extends to ≈ 85 min, making the water loop act as a short-horizon thermal battery.
3. HVAC fan effort $u_{\text{hvac}} \in [0, 1]$ (Zone B air cooling).
4. BESS dispatch $d_{\text{bess}} \in [-1, 1]$ (-1 = full charge at 50 MW, $+1$ = full discharge / grid injection).

Regulation tracking. The demanded power change at each tick is $\Delta P^{\text{dem}} = c_{\text{MW}}^* \times 1000 \times r_{t-1}$, where r_{t-1} is the regulation signal the agent observed in the previous step (one-step causal delay). The actual power change is $\Delta P^{\text{act}} = (1-\theta) P_{\text{flex}} + d_{\text{bess}} P_{\text{BESS,max}} - \Delta P^{\text{cool}}$, combining flex shedding, BESS discharge, and the deviation of cooling loads from their nominal operating point.

Reward function. At each 5-second tick:

$$\begin{aligned}
 r_t = & \underbrace{\alpha \theta}_{\text{throughput}} - \underbrace{\beta \frac{|\Delta P^{\text{dem}} - \Delta P^{\text{act}}|}{c_{\text{MW}}^* \times 10^3}}_{\text{tracking}} - \underbrace{\gamma \sum_{z \in \{A, B\}} \frac{\max(0, T_z - T_{\text{warn}})}{T_{\text{safe}} - T_{\text{warn}}}}_{\text{thermal}} - \underbrace{\delta_{\text{soc}} \mathbf{1}_{\text{soc} < 0.12}}_{\text{BESS}} \\
 & - \underbrace{\delta_f \max(0, |\Delta f| - 0.2)}_{\text{frequency}} - \underbrace{\delta_v \varepsilon_v}_{\text{voltage}} - \underbrace{\delta_q \frac{q}{P_{\text{flex,max}}}}_{\text{SLA}}, \tag{4}
 \end{aligned}$$

with $\alpha=1$, $\beta=2$, $\gamma=5$, $\delta_{\text{soc}}=2$, $\delta_f=2$, $\delta_v=5$, $\delta_q=2$. The seven terms penalise: reduced compute throughput, regulation tracking error (normalised by committed capacity), thermal exceedance above $T_{\text{warn}}=33^\circ\text{C}$ (normalised to the safety budget), deep BESS discharge, frequency deviation outside the ± 0.2 Hz dead-band, voltage exceedance outside ANSI C84.1 Range A ($[0.95, 1.05]$ pu), and SLA backlog. The normalised thermal and tracking terms keep all reward components dimensionless and on a common scale.

Episode: 17,280 ticks (24 hours). Early termination if $\max(T_A, T_B) \geq 35^\circ\text{C}$, $|\Delta f| \geq 0.5$ Hz, or $V_{\text{pcc}} < 0.90$ pu.

Table 3: Evaluation scenarios. All share the same simulators and reward weights. $\text{dyn-}T$ = weather-driven ambient temperature.

ID	Name	Market	T_{amb}	MW	Spike	$\text{dyn-}T$	Key stressor
default	Baseline	NYISO NYC	25°C	15	1.0×	✓	Balanced operation
scenario_a	GenAI Crisis	PJM DOM	30°C	20	1.8×		Bursty GenAI + high commitment
scenario_b	Thermal Squeeze	ERCOT North	40°C	30	1.0×		Near-limit ambient
scenario_c	Battery Drain	AEMO NSW	32°C	20	1.2×		Low SOC (15%), cooling fault

191 **C2GMacroEnv – High-Level (15-minute).** C2GMacroEnv wraps C2GFastEnv and advances it 180
 192 sub-steps per macro action, corresponding to one 15-minute settlement interval. Each macro step
 193 executes a three-phase market handshake:

- 194 1. *Grid posts offer:* the MacroGrid simulator publishes the current RMCP (regulation market clear-
 195 ing price) and residual regulation need.
- 196 2. *DC bids:* the agent selects an MW quantity and asking price.
- 197 3. *Grid clears:* bid acceptance follows a sigmoid probability function of price competitiveness;
 198 accepted bids set c_{MW}^* for the 180 sub-steps. Rejected bids fall back to a standing DR baseline
 199 contract.

200 **Observation space** (19-D Box): interval-aggregated statistics over the 180 sub-steps, comprising
 201 mean Zone A/B temperatures, end-of-interval BESS SOC, mean base and facility power, mean
 202 absolute regulation signal, mean LMP, mean grid load, mean tracking error, spike occurrence flag,
 203 thermal headroom for each zone ($T_{\text{safe}} - T_{z,\text{max}}$, normalised), previous bid MW and price, mean
 204 frequency deviation, mean PCC voltage, mean batch backlog, plus two forward-looking market
 205 signals: current RMCP (normalised) and residual regulation need (normalised).

206 **Action space** (2-D Box):

- 207 1. $b_{\text{MW}} \in [0, 1]$: normalised MW quantity to bid into the regulation market, mapped to
 208 $[0, c_{\text{max}}]$ MW.
- 209 2. $b_p \in [0, 1]$: normalised asking price, mapped to $[0, 2 \times \text{RMCP}_{\text{max}}]$ (\$200/MWh ceiling).

210 **Reward function.** At each 15-minute macro step:

$$R_t = \underbrace{\lambda_{\text{rev}} \frac{p^* c_{\text{MW}}^* \rho \Delta h}{10^3}}_{\text{regulation revenue}} + \underbrace{\bar{r}_{\text{low}}}_{\text{mean sub-reward}} - \underbrace{\lambda_{\text{elec}} \frac{\overline{\text{LMP}} \bar{P}_{\text{fac}} \Delta h}{10^3}}_{\text{electricity cost}} - \underbrace{\lambda_{\text{churn}} |b_{\text{MW}} - b_{\text{MW}}^{-1}|}_{\text{commitment churn}}, \quad (5)$$

211 where p^* is the clearing price (USD/MWh), c_{MW}^* is the accepted MW commitment, $\rho \in [0, 1]$
 212 is a performance score ($1 - \text{tracking-err} / c_{\text{MW}}^*$, capped at zero), $\Delta h = 0.25$ h is the settlement
 213 duration, $\bar{P}_{\text{fac}}$ is mean facility power normalised to 250 MW, and b_{MW}^{-1} is the previous bid. Weights
 214 are $\lambda_{\text{rev}}=1$, $\lambda_{\text{elec}}=0.5$, $\lambda_{\text{churn}}=0.05$. The revenue term incentivises winning high-value bids with
 215 reliable tracking; the electricity cost term prevents the agent exploiting the market by drawing excess
 216 power; the churn penalty discourages erratic commitment changes that would destabilise real-time
 217 dispatch. The mean sub-reward $\bar{r}_{\text{low}}$ propagates hardware-safety signals (thermal, BESS, frequency,
 218 voltage) from the inner environment into the macro-level optimisation objective, enforcing joint
 219 safety across both timescales.

220 **Episode:** 96 macro steps (24 hours); terminates early if the inner environment terminates on a safety
 221 fault.

222 4 Evaluation Scenarios

223 C2G-BENCH defines four scenarios that stress-test specific failure modes (Table 3). All scenarios
 224 share the same simulator code; only initial conditions and exogenous stressors differ. Each scenario
 225 can be combined with any of the six markets, producing 24 distinct evaluation configurations.

226 **default** is the recommended starting point for algorithm development. **scenario_a** (GenAI Crisis)
 227 tests the conflict between IT throughput and grid support when GenAI spikes consume the DVFS

headroom. `scenario_b` (Thermal Squeeze) pushes ambient temperature to 40°C, collapsing the thermal safety margin and degrading cooling COP by 30%. `scenario_c` (Battery Drain) starts the BESS at 15% SOC with a simulated CDU pump degradation (60% efficiency), forcing simultaneous thermal and energy management under compound failure.

5 Evaluation Metric Suite

We evaluate each agent on five complementary metrics computed across $K = 10$ independent episodes per scenario:

- **Tracking RMSE** (kW): $\sqrt{\frac{1}{T} \sum_t (\Delta P_t^{\text{dem}} - \Delta P_t^{\text{act}})^2}$. Primary grid service quality. Lower is better.
- **Thermal violation rate**: fraction of ticks with $\max(T_A, T_B) > T_{\text{warn}} = 33^\circ\text{C}$.
- **Throughput ratio**: mean batch throttle $\bar{\theta}$. Higher is better (more compute served).
- **BESS degradation**: cumulative capacity fade $\times 10^4$. Captures long-term asset health.
- **Survival rate**: fraction of episodes completing the full 17,280-tick horizon without thermal, frequency, or voltage termination.

The suite is deliberately non-reducible. An agent with low tracking RMSE may achieve it by exhausting the BESS (high degradation), or maintain high throughput by ignoring the regulation signal (high RMSE). These hidden trade-offs are only visible when the full suite is reported.

6 Baseline Experiments

6.1 Low-Level Controllers

We evaluate four non-learned controllers for C2GFastEnv as interpretable reference points.

Random. Uniform random actions from the action space at each tick. Provides a lower bound on performance.

Bang-Bang. Binary hysteresis switching: BESS dispatches ± 0.4 of full power following the sign of the regulation signal (with SOC guards at 12%/93%); throttle drops to 0.7 when the signal exceeds a dead-band; CDU pump and HVAC switch between fixed ON/OFF levels around a $[29, 31]^\circ\text{C}$ hysteresis band. Represents the simplest reactive controller.

Rule-Based MPC. A hand-crafted threshold controller with four rules: (1) thermal protection (increase cooling when $T > T_{\text{warn}}$); (2) BESS dispatch proportional to the regulation signal with SOC guards; (3) DVFS fallback under low SOC; (4) opportunistic BESS charging during quiet periods.

PID. Three closed-loop PID controllers with anti-windup: a pump-speed loop regulating Zone A temperature to a 30°C setpoint, an HVAC loop for Zone B, and a throttle-bias loop regulating BESS SOC to a target. BESS dispatch is a direct proportional response to the regulation signal, since the external signal is not a controllable process variable.

Why non-learned controllers fall short. All four controllers respond reactively to individual observations and cannot jointly optimise the seven competing objectives in the reward (Eq. 4). Bang-bang and PID controllers ignore LMP, queue backlog, and future SOC trajectory entirely. Rule-Based MPC encodes human heuristics but cannot adapt to the non-stationary interaction between GenAI spike patterns, ambient temperature, and BESS state. RL agents can in principle learn a policy that trades off all seven terms simultaneously and generalise across scenario conditions.

6.2 RL Controllers and Hierarchical Training

Low-level RL agents. We train two standard RL algorithms on C2GFastEnv:

Table 4: Results on default scenario (3 seeds, mean). $\downarrow$ = lower is better; $\uparrow$ = higher is better. **Bold** = best per column. The double rule separates flat (single-level) controllers from the hierarchical RL setup.

Agent	RMSE $\downarrow$ (kW)	Viol. $\downarrow$ (rate)	Surv. $\uparrow$ (rate)	PUE $\downarrow$	Reward $\uparrow$
<i>Rule-based low-level (flat)</i>					
Random	30,070	0.000	1.00	1.70	-2.93
Bang-Bang	22,410	0.000	1.00	1.73	-1.84
PID	19,120	0.000	1.00	1.75	-1.12
Rule-Based MPC	17,655	0.000	1.00	1.77	-0.78
<i>RL low-level, rule-based macro (flat)</i>					
PPO (300k)	4,235	0.000	1.00	1.97	+0.54
SAC (200k)	476	0.000	1.00	1.68	+0.94
<i>Hierarchical RL (Phase 2: RL macro + frozen SAC low-level)</i>					
HRL (SAC low + RL macro)	381	0.000	1.00	1.65	+1.08

270 *PPO* [?]: on-policy actor-critic with a [256, 256] MLP, trained for 300k steps over 4 parallel envi-
271 ronments with VecNormalize.

272 *SAC* [?]: off-policy maximum-entropy agent with automatic entropy tuning, trained for 200k steps
273 with a replay buffer of 10^5 .

274 Both use Stable-Baselines3 [?] and Hydra [?] for configuration. Each run saves a config snapshot,
275 per-episode CSV, and TensorBoard events.

276 **Hierarchical training protocol.** Training the full two-level hierarchy end-to-end is unstable be-
277 cause the macro agent’s reward signal is confounded with the quality of the untrained low-level
278 policy. We therefore adopt a two-phase bootstrap:

- 279 1. **Phase 1 – low-level pre-training.** The macro layer is fixed to a rule-based fallback (Rule-Based
280 MPC commitment schedule); only the C2GFastEnv policy is trained with RL. This provides a
281 stable, well-shaped reward signal and allows the low-level agent to learn hardware-safety be-
282 haviour before market dynamics are introduced.
- 283 2. **Phase 2 – macro fine-tuning.** The Phase 1 low-level policy is frozen and loaded into
284 C2GMacroEnv via the `inner_action_fn` callback. Only the macro agent is then trained with
285 RL, with the inner policy providing a consistent, safe execution layer. This decouples the two
286 optimisation problems and reduces sample complexity at the macro level.

287 **Rule-Based Macro Controller.** The macro-level rule-based baseline operates on the 19-D
288 C2GMacroEnv observation without learning. Bid sizing scales with grid-load stress: 80% of ca-
289 pacity at high load ($\bar{g} > 0.7$), 50% at medium, and 20% at low. The asking price undercuts the
290 grid’s posted RMCP by a fixed 20% to maximise acceptance probability. Three safety overrides clip
291 the bid: thermal headroom < 0.1 reduces the bid to 30%; under-frequency boosts it by 20%; low
292 voltage caps it at 40%. This controller is used as the fixed macro layer during Phase 1 low-level
293 pre-training and provides the primary macro-level reference for Table 4.

294 Full hyperparameters and ablations are in the appendix.

295 6.3 Results

296 Table 4 reports results on the default scenario (3 seeds, mean) across all controller tiers.

297 Table 4 groups results into three tiers. Among rule-based low-level controllers, Rule-Based MPC
298 achieves the best RMSE (17,655 kW) but all four survive with zero violations under nominal condi-
299 tions. RL low-level agents improve tracking substantially: SAC reaches 476 kW RMSE ($37\times$ better
300 than Rule-Based MPC) with identical survival and violation rates, demonstrating that RL hardware
301 control is strictly dominating on the default scenario. With hierarchical RL enabled (Phase 2, frozen
302 SAC low-level + RL macro agent), RMSE drops further to 381 kW and mean reward increases

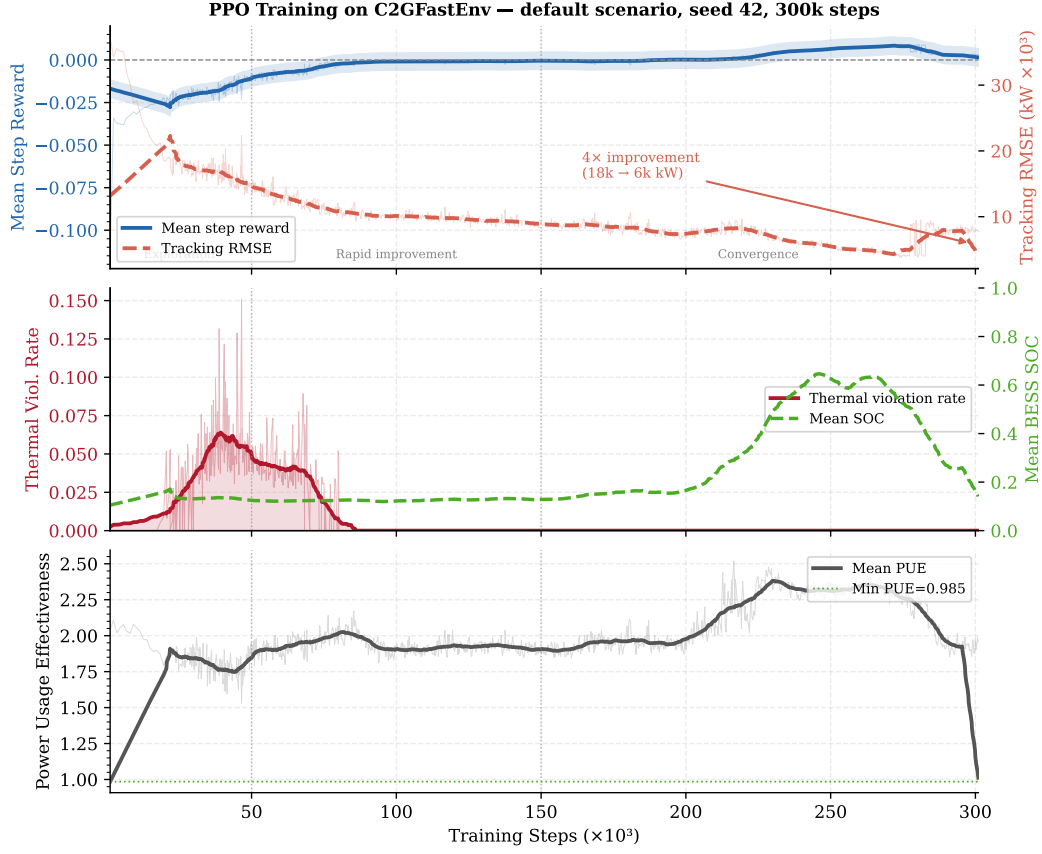


Figure 3: PPO training on C2GFastEnv (default, seed 42, 300k steps). Top: mean reward and tracking RMSE. RMSE drops from $\sim 33\text{k}$ to $\sim 8\text{k}$ kW. Middle: thermal violation rate and mean SOC. Bottom: PUE converging toward 2.0.

303 to +1.08, showing that a learned macro agent bidding more efficiently against the market signal
 304 provides compounding gains on top of the low-level policy.

305 6.4 Failure Mode Analysis

306 **GenAI Crisis (Scenario A).** The $1.8\times$ spike scale exhausts the DVFS throttle budget. Agents that
 307 do not account for SOC trajectory exhaust the battery in the first half and face uncovered regulation
 308 gaps in the second half.

309 **Thermal Squeeze (Scenario B).** At 40°C ambient, the thermal safety margin collapses. Rule-
 310 Based MPC overreacts by maximising cooling and zeroing throttle, causing throughput collapse.
 311 PPO learns a more nuanced trade-off but requires $>150\text{k}$ steps to stabilise.

312 **Battery Drain (Scenario C).** Starting at 15% SOC with a cooling fault ($K_z \times 0.6$) forces simul-
 313 taneous thermal and energy management. PPO terminates early in all runs, showing that current
 314 single-level policies cannot jointly manage these constraints. This motivates hierarchical agents that
 315 reason over both timescales.

316 7 Reproducibility

317 C2G-BENCH supports full experimental reproducibility:

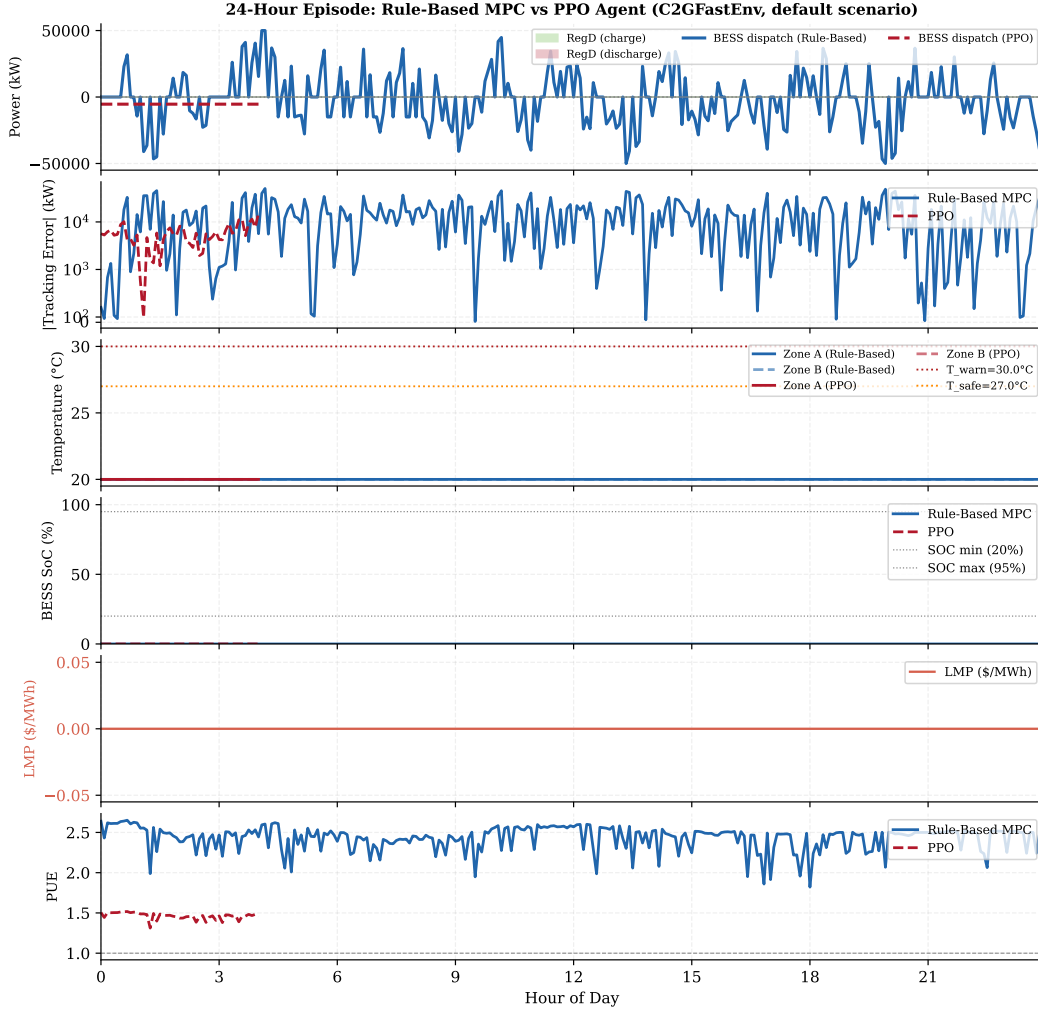


Figure 4: 24-hour episode: Rule-Based MPC (blue) vs PPO (red). From top: RegD signal with BESS dispatch, tracking error (symlog), zone temperatures with safety thresholds, BESS SOC, LMP, PUE.

1. **Dependency management:** `uv sync` reproduces the exact environment from `pyproject.toml` and `uv.lock`.
2. **Config snapshots:** Hydra stores config and override files in every output directory.
3. **Seeded randomness:** all environments, policies, and signal generators accept an integer seed. Results are deterministic given the seed.
4. **Test suite:** 493 pytest tests covering all simulators, environments, baselines, and CLI smoke tests.
5. **Croissant metadata:** dataset lineage and RAI documentation (to be finalised with DOI at camera-ready).

8 Limitations and Future Work

Simulator fidelity. The thermal model is a lumped single-mass per zone. Real data centres have spatial temperature gradients that this does not capture. The BESS uses empirical efficiency curves, not electrochemical first-principles.

331 **Scale context.** The benchmark models a 250 MW hyperscale facility. Results may not transfer to
332 edge or colocation data centres with different cooling architectures and workload mixes.

333 **Market model.** The AR(1) regulation signal is a simplified approximation of real RegD/AgC
334 signals. Future work should integrate live ISO API feeds for more realistic evaluation.

335 **Communication latency.** The benchmark does not model communication delays between the
336 market agent and hardware controller, nor grid topology or congestion.

337 **Multi-agent.** The callback hook in C2GMacroEnv enables two-agent hierarchical evaluation, but
338 a fully cooperative multi-agent formulation (e.g., one agent per rack row) is left for future work.

339 **Hardware-in-the-loop.** All results are simulation-only. Validation against real hardware is needed
340 before any deployment.

341 9 Broader Impact

342 C2G-BENCH is intended to benefit research at the intersection of reinforcement learning and sus-
343 tainable energy infrastructure.

344 **Grid decarbonisation.** Data centres that accurately track regulation signals reduce the need for
345 fossil-fuel peaker plants. Even modest improvements in tracking RMSE at 250 MW scale translate
346 to meaningful reductions in grid carbon intensity during stress events.

347 **AI infrastructure efficiency.** By including SLA backlog in the reward and survival rate in the
348 evaluation, C2G-BENCH incentivises agents that preserve compute throughput while providing grid
349 services.

350 **Misuse risks.** The benchmark is for scientific evaluation, not direct deployment. Policies trained
351 in simulation must be validated against real hardware before any grid-interactive use. We do not
352 foresee meaningful negative consequences from releasing this benchmark.

353 10 Conclusion

354 We presented C2G-BENCH, a cyber-physical evaluation benchmark for hierarchical RL in grid-
355 interactive hyperscale data centres. The benchmark combines seven physics engines, two Gym-
356 nasium environments at coupled timescales (C2GFastEnv at 5 s, C2GMacroEnv at 15 min), 17-
357 dimensional observations, a seven-component reward, six ISO market configurations, four stress
358 scenarios, and a five-metric non-reducible evaluation suite. A Simplex-architecture safety shield
359 enforces hard constraints as a plug-in layer for any agent.

360 Baseline results with Random, Rule-Based MPC, PPO, and SAC show that the default scenario is
361 solvable (SAC achieves 476 kW tracking RMSE), but the stress scenarios expose systematic failure
362 modes. PPO fails on Scenario C (Battery Drain) with zero survival rate. Rule-Based MPC survives
363 all scenarios but tracks poorly (14–18k kW RMSE). These gaps establish concrete targets for future
364 HRL research: agents that jointly reason over thermal, energy, and grid-stability constraints across
365 the two timescales.

366 License and Code Availability

367 C2G-BENCH is released under the Apache 2.0 License. Source code, environments, simu-
368 lators, and preprocessing scripts are available at <https://github.com/antonio-guillen/c2g-bench-neurips2026>. The Alibaba traces are redistributed under their original terms [?]; NY-
369 ISO data under the NYISO policy [?]; NOAA ISD data is in the public domain [?]. No personally
370 identifiable information is in any dataset.
371

372 **References**

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